6.6 Parametric Equations

Pre-Calculus 6.6 · Student sheet

Name _____ Period ____ Date _____

I can…

- sketch a curve from parametric equations and show its orientation.
- eliminate the parameter.
- write parametric equations for a graph.

Vocabulary · fill in the missing word

Plane curve — the set of points (x, y) = (f(t), g(t)) for t in an _____

Parametric equations — x = _____ = g(t)

Parameter — the third _____ t

Orientation — the _____ the curve is traced as t increases

Part 1 · Time and position

Fill in $x = t^2$ and $y = t/2$. Plot the points in order and draw arrows.

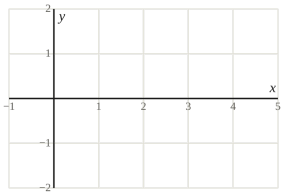
t	−2	−1	0	1	2
x					
y					

Part 2 · Notes and examples (fill in each note, then work the example)

Sketching: make a table of t values, plot (x, y) in order of increasing t, and use arrows to show the _____.

Example 1

Sketch $x = t^2$, $y = \frac{t}{2}$ for $-2 \le t \le 2$. Show the orientation.



Eliminating the parameter: solve one equation for t and _____ into the other.

Example 2

Eliminate the parameter.

a. $x = t - 1$, $y = 2t + 1$

b. $x = t^2$, $y = \frac{t}{2}$

Trig parameters: solve for cos θ and sin θ, then use _____.

Example 3

Eliminate the parameter: $x = 3\cos\theta$, $y = 4\sin\theta$. Identify the curve.

Example 4

Write parametric equations for $y = 1 - x^2$ using (a) $t = x$ and (b) $t = 1 - x$.

Example 5

A ball is hit 3 feet above the ground at 88 ft/s at an angle of 35°: $x = (88\cos 35°)t$, $y = 3 + (88\sin 35°)t - 16t^2$. About how far does it travel before hitting the ground?

Part 3 · Partner practice

Partner 1

Eliminate the parameter: $x = 2t + 1$, $y = t - 3$.

Partner 2

Eliminate the parameter: $x = 2\cos\theta$, $y = 2\sin\theta$.